

Transformations of $f(x) = |x|$, Part I

TEACHER KEY | Algebra II Honors | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe, graph, and justify the effect of a and b on $f(x) = a|bx|$, including why some transformations produce no visible change.

Vocabulary

Vertical stretch/compression — $af(x)$; $|a| > 1$ stretches away from the x-axis, $0 < |a| < 1$ compresses toward it.

Horizontal stretch/compression — $f(bx)$; $|b| > 1$ compresses toward the y-axis, $0 < |b| < 1$ stretches away from it.

Invariant point — a point that does NOT move under a transformation. For $f(x) = a|bx|$, the **vertex** is always invariant.

Part 1 — The Effect of a and b, Combined

Function	a	b	Effect
$f(x) = 2 x $	2	1	vertical stretch x2
$f(x) = (1/2) x $	1/2	1	vertical compression
$f(x) = 2x $	1	2	horizontal compression
$f(x) = -2 3x $	-2	3	stretch x2, compress x3, reflect over x-axis

Part 2 — Why the Vertex Never Moves

Show algebraically that (0, 0) is on the graph of $f(x) = a|bx|$ no matter what a and b are:

$$f(0) = a|b \cdot 0| = a|0| = a \cdot 0 = 0 \rightarrow \text{the vertex (0, 0) is invariant under every a and b}$$

Part 3 — Why Reflecting Over the y-axis Does Nothing

$f(x) = |-x|$ should reflect the parent over the y-axis ($b = -1$). Prove algebraically that it produces the SAME graph as $f(x) = |x|$:

$$|-x| = |-1| \cdot |x| = 1 \cdot |x| = |x| \rightarrow \text{identical to the parent - this is the same even-function fact from Day 1 (f(-x) = f(x))}$$

Part 4 — Which Graphs Are Really the Same?

For each pair, determine whether the two functions produce the SAME graph. Justify using the properties of absolute value.

- $f(x) = 3|x|$ and $f(x) = |3x|$ **SAME** since $|3x| = |3||x| = 3|x|$
- $f(x) = -2|x|$ and $f(x) = 2|-x|$ **DIFFERENT** $-2|x|$ opens downward; $2|-x| = 2|x|$ opens upward
- $f(x) = (1/2)|4x|$ and $f(x) = 2|x|$ **SAME** since $(1/2)|4x| = (1/2)(4)|x| = 2|x|$
- $f(x) = |6x|$ and $f(x) = 6|x|$ **SAME** since $|6x| = |6||x| = 6|x|$

Part 5 — Going Further: Factoring Out a Constant

Tomorrow's lesson adds shifts: $f(x) = a|x - c| + d$. Some functions LOOK like they have a b-value inside, but a constant can be factored out to reveal the real a. Try it:

$$f(x) = 2|3x - 6| = 2|3(x - 2)| = 2 \cdot 3 \cdot |x - 2| = 6|x - 2| \rightarrow \text{the REAL a-value is 6}$$

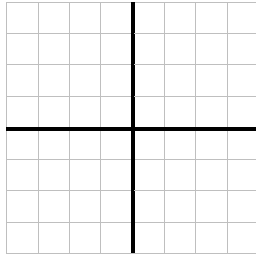
You Try. Factor $f(x) = 4|2x + 8|$ to find the real a-value and the vertex's x-coordinate.

$$f(x) = 4|2(x + 4)| = 8|x + 4| \rightarrow a = 8 \text{ vertex x-coordinate } -4$$

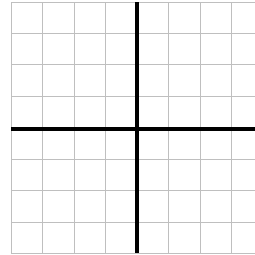
Part 6 — Graph and Compare

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = 2|x|$



B. $f(x) = (1/2)|x|$



Part 7 — Practice

- Factor $f(x) = 3|6x + 18|$ into $a|x - c|$ form and give the real a-value. **$18|x + 3|$, $a = 18$**
- Order from flattest to steepest: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$. **$(1/4)|x|$, $|x|$, $-2|x|$, $3|x|$**
- True or False: $f(x) = |-5x|$ and $f(x) = |5x|$ produce different graphs. Justify. **False - $|-5x| = |-5||x| = 5|x| = |5x|$, identical**
- Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look identical for every x. What relationship must a and b satisfy? **$a = |b|$**
- Factor $f(x) = 7|-2x + 8|$ into $a|x - c|$ form and give the real a-value and vertex x-coordinate. **$14|x - 4|$, $a = 14$, x-coordinate 4**

Exit Ticket

Are $f(x) = -3|x|$ and $f(x) = 3|-x|$ the same graph? Justify algebraically, then describe every transformation present. Finally, factor $f(x) = 5|4x - 20|$ to find its real a-value.

No $-3|-x| = 3|x|$ opens upward, while $-3|x|$ opens downward (both stretch by 3). $5|4x-20| = 5|4(x-5)| = 20|x-5|$, so $a = 20$